Functions in MySQL – Exercise 17

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Program 1 – Function that returns the level of a customer based on credit limit

1. USE THE DATABASE AND CREATE THE CUSTOMERS TABLE

```
USE CollegeDB;
DROP TABLE IF EXISTS customers;
CREATE TABLE customers (customerNumber INT PRIMARY KEY, customerName
VARCHAR(50), creditLimit DECIMAL(10,2));

INSERT INTO customers VALUES
(101, 'Kumar Traders', 21000.00),
(102, 'Sri Gifts', 71800.00),
(103, 'Bala Stores', 0.00),
(104, 'Lakshmi Mart', 9500.00),
(105, 'Anand and Co', 50000.00);
```

Output:

```
Database changed

Query OK, 0 rows affected (0.048 sec)

Query OK, 0 rows affected (0.096 sec)

Query OK, 5 rows affected (0.010 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

2. CREATE THE FUNCTION CustomerLevel

```
DELIMITER $$
CREATE FUNCTION CustomerLevel(p_creditLimit DECIMAL(10,2))
RETURNS VARCHAR(20)
DETERMINISTIC
BEGIN
    DECLARE customerLevel VARCHAR(20);

    IF p_creditLimit > 50000 THEN
        SET customerLevel = 'PLATINUM';
    ELSEIF p_creditLimit >= 10000 AND p_creditLimit <= 50000 THEN
        SET customerLevel = 'GOLD';
    ELSEIF p_creditLimit < 10000 THEN
        SET customerLevel = 'SILVER';
    END IF;

    RETURN customerLevel;
END$$
DELIMITER ;
```

Output:

```
Query OK, 0 rows affected (0.023 sec)
```

3. CALL THE FUNCTION WITH A VALUE

SELECT CustomerLevel(60000);

Output:

```
+----------------------+
| CustomerLevel(60000) |
+----------------------+
| PLATINUM             |
+----------------------+
1 row in set (0.001 sec)
```

4. USE THE FUNCTION ON THE CUSTOMERS TABLE

SELECT customerName, creditLimit, CustomerLevel(creditLimit) AS customerLevel
FROM customers
ORDER BY customerNumber;

Output:

```
+---------------+-------------+---------------+
| customerName  | creditLimit | customerLevel |
+---------------+-------------+---------------+
| Kumar Traders |    21000.00 | GOLD          |
| Sri Gifts     |    71800.00 | PLATINUM      |
| Bala Stores   |        0.00 | SILVER        |
| Lakshmi Mart  |     9500.00 | SILVER        |
| Anand and Co  |    50000.00 | GOLD          |
+---------------+-------------+---------------+
5 rows in set (0.003 sec)
```

Program 2 – Recursive procedure to compute the factorial of a number

5. SET THE RECURSION DEPTH

   SET @@GLOBAL.max_sp_recursion_depth = 255;

   SET @@session.max_sp_recursion_depth = 255;

   Output:

```
Query OK, 0 rows affected (0.000 sec)

Query OK, 0 rows affected (0.000 sec)
```

6. CREATE THE RECURSIVE PROCEDURE Factorial

```
DELIMITER $$
CREATE PROCEDURE Factorial(IN n INT, OUT result BIGINT)
BEGIN
    DECLARE prev BIGINT;

    IF n <= 1 THEN
        SET result = 1;
    ELSE
        CALL Factorial(n - 1, prev);
        SET result = n * prev;
    END IF;
END$$
DELIMITER ;
```

   Output:

```
Query OK, 0 rows affected (0.017 sec)
```

7. FACTORIAL OF 5

   CALL Factorial(5, @f);

   SELECT @f AS factorial_of_5;

   Output:

```
Query OK, 0 rows affected (0.002 sec)

+----------------+
| factorial_of_5 |
+----------------+
|            120 |
+----------------+
1 row in set (0.000 sec)
```

8. FACTORIAL OF 7

```
CALL Factorial(7, @f);
SELECT @f AS factorial_of_7;
```

Output:

```
Query OK, 0 rows affected (0.005 sec)

+----------------+
| factorial_of_7 |
+----------------+
|           5040 |
+----------------+
1 row in set (0.000 sec)
```

9. FACTORIAL OF 10

```
CALL Factorial(10, @f);
SELECT @f AS factorial_of_10;
```

Output:

```
Query OK, 0 rows affected (0.004 sec)

+-----------------+
| factorial_of_10 |
+-----------------+
|         3628800 |
+-----------------+
1 row in set (0.000 sec)
```